nrmk AI



Master-Slave Teleoperation System for Indy Robot

Lee Hyesung

Internship Wrap-up Presentation

Feb 27, 2025

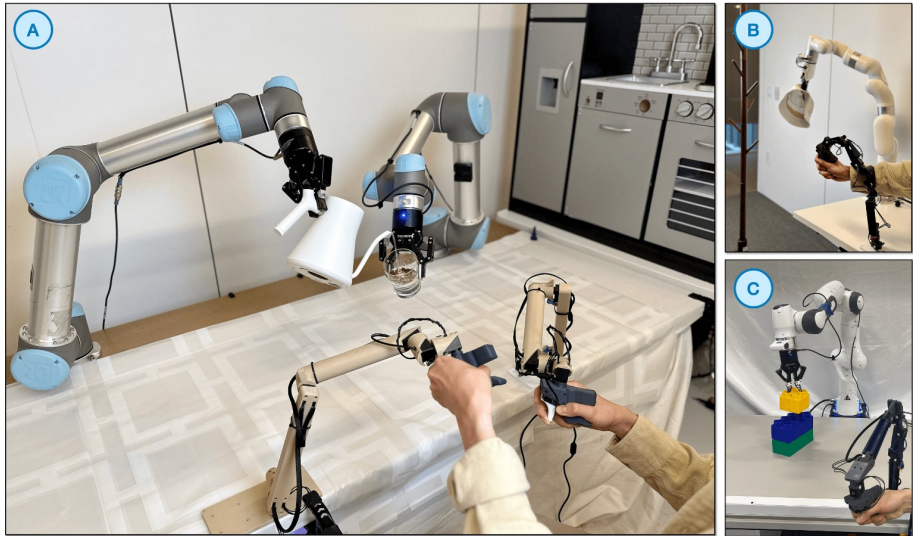
POSITION TRACKING

GELLO: A General, Low-Cost, and Intuitive Teleoperation Framework for Robot Manipulators

Previous Researches

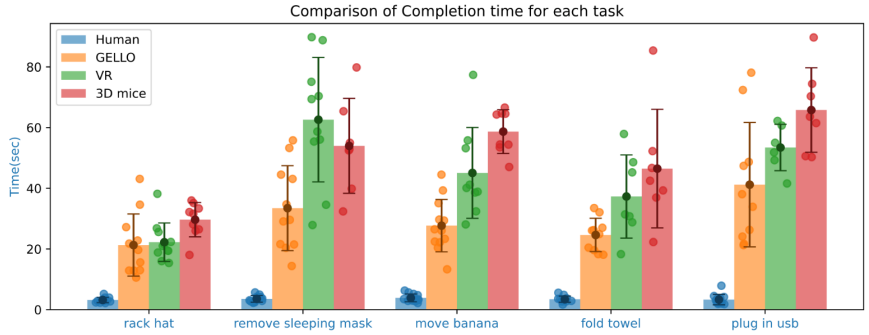
Teleoperation device (Joint level control)

- Intuitive control method rather than VR, 3D Mice
- Sense the kinematic constraints directly
- Extreme low cost controller



Demonstration

- Refill water bottle, grab drink, make ramen, make toast, erase board, open laptop



Shortcoming

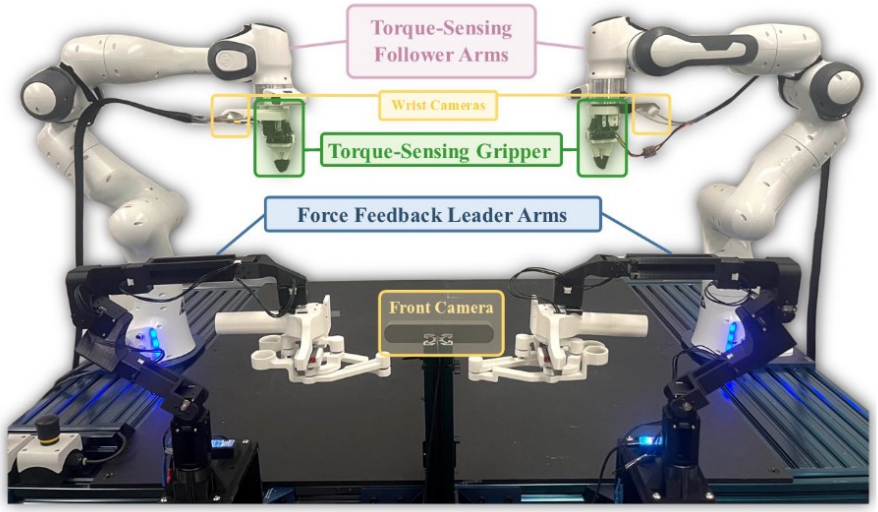
- No User feedback
- Indifferent feelings due to mechanical gravity compensation

FACTR: Force-Attending Curriculum Training for Contact-Rich Policy Learning

Previous Researches

Teleoperation device with control

- Force feedback (mirror joint torque)
- Regulate the redundancy (7 dof robot)
- Gravity compensation
- Friction compensation
- Learning with the transformer structure

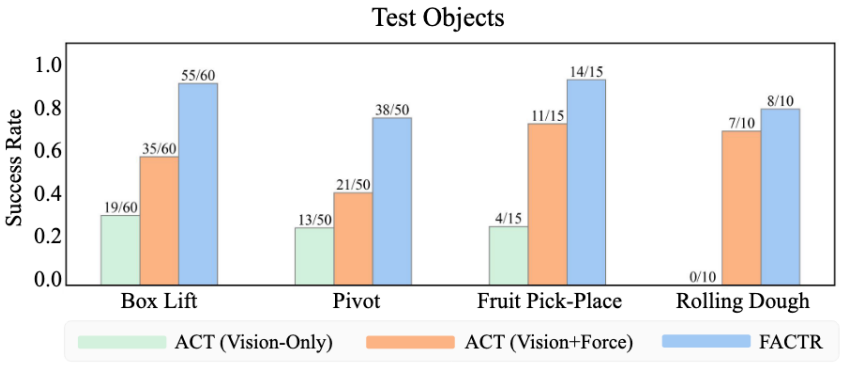


Demonstration

- box lift, pivot, fruit pick and place, rolling dough

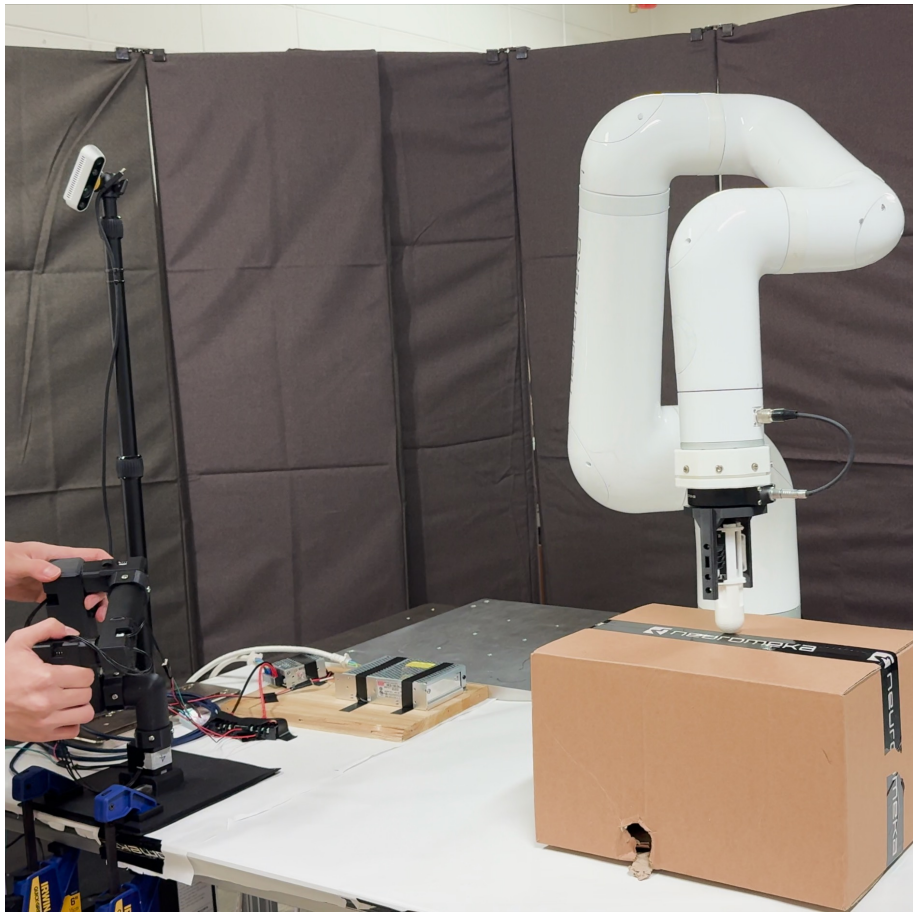
Shortcoming

- Applicable only to the joint torque sensible / compliance control robot
- No demonstration for hard contact



Why the Master Device with Feedback / Compliance is Needed?

The worth of this project



Force Feedback:

Sense the environmental interaction on the surface

>>> Precise control with the sense of touch

Compliance:

Safe control despite of the rigid robot arm

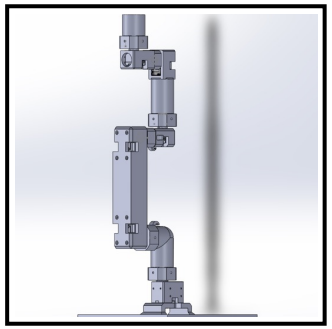
>>> joint command can penetrate through the ground / momentum is absorbed

>>> Safe and less stressful manipulation

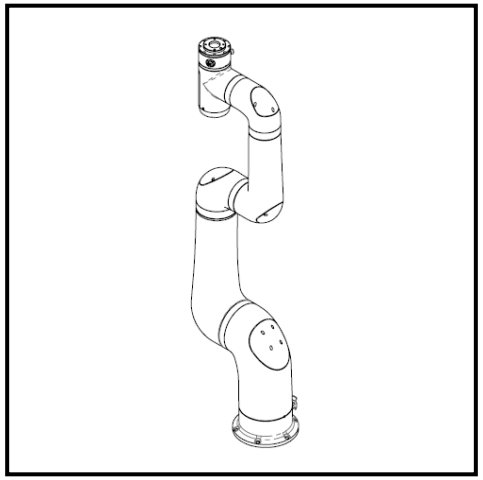
System Flow

Base Structure

Joint position (q)



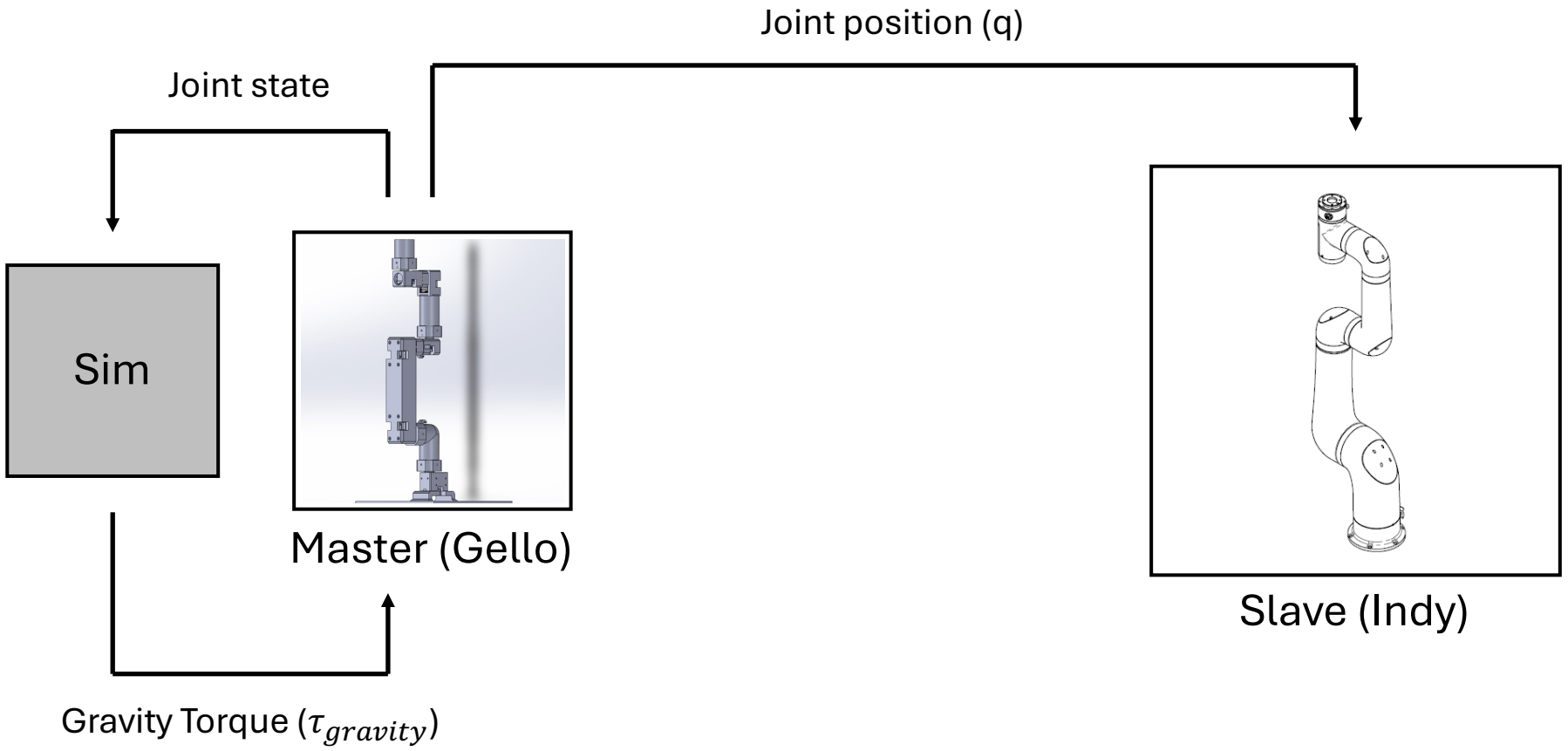
Master (Gello)



Slave (Indy)

System Flow

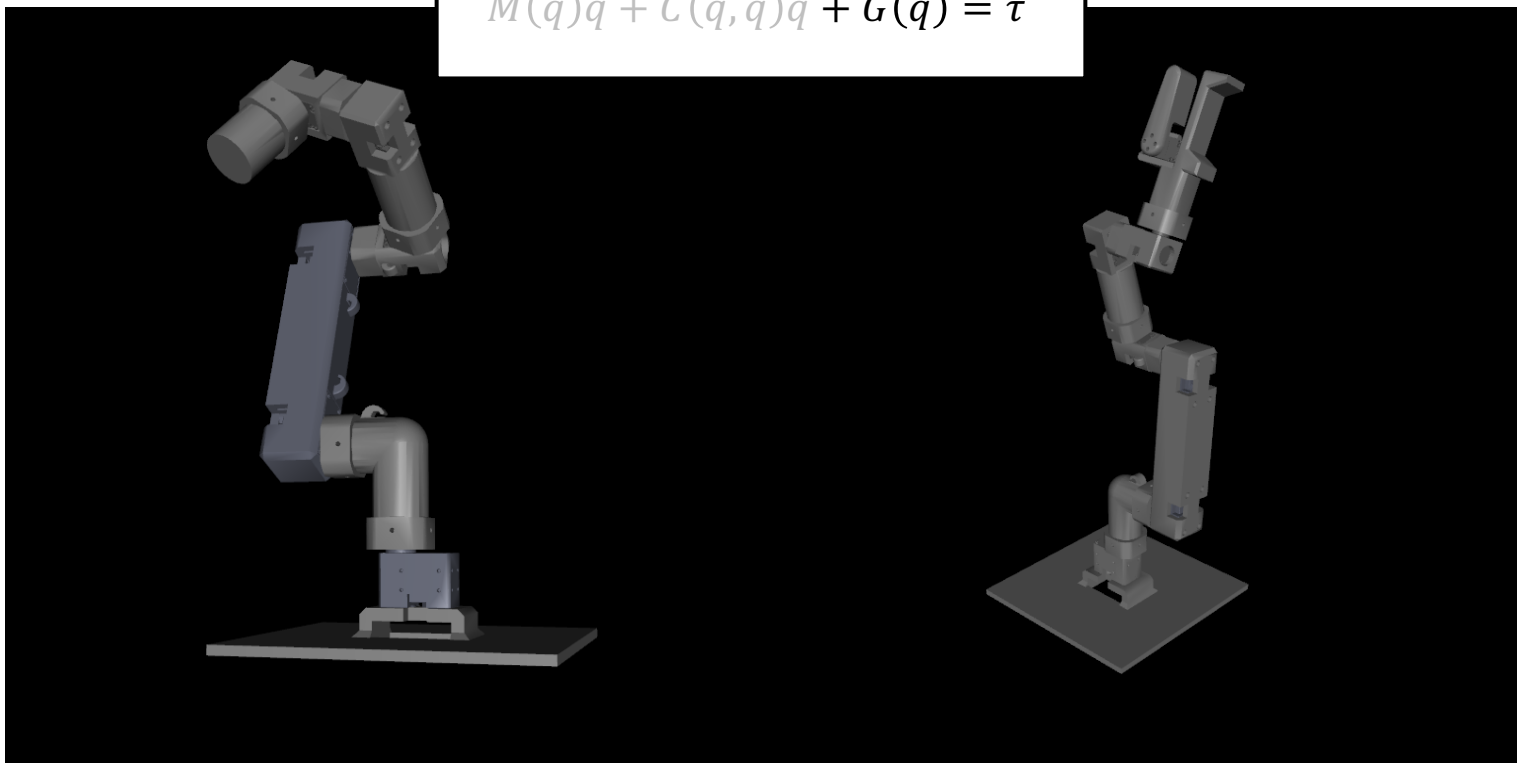
Gravity Compensation



System Flow

Gravity Compensation

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) = \tau$$



What is running under the simulation...

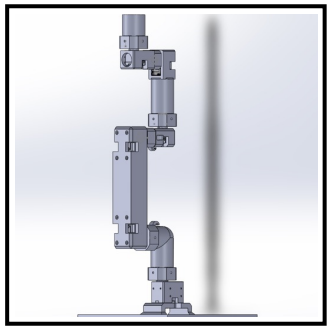
System Flow

Force Feedback

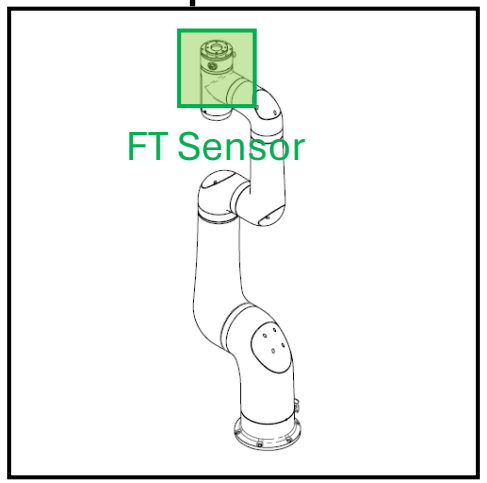
Joint position (q)

Joint state + External wrench

Sim



Master (Gello)



Slave (Indy)

Gravity Torque ($\tau_{gravity}$)

+ External Torque (τ_{ext})

$$\tau_{ext} = \mathcal{J}^T \mathcal{F}$$

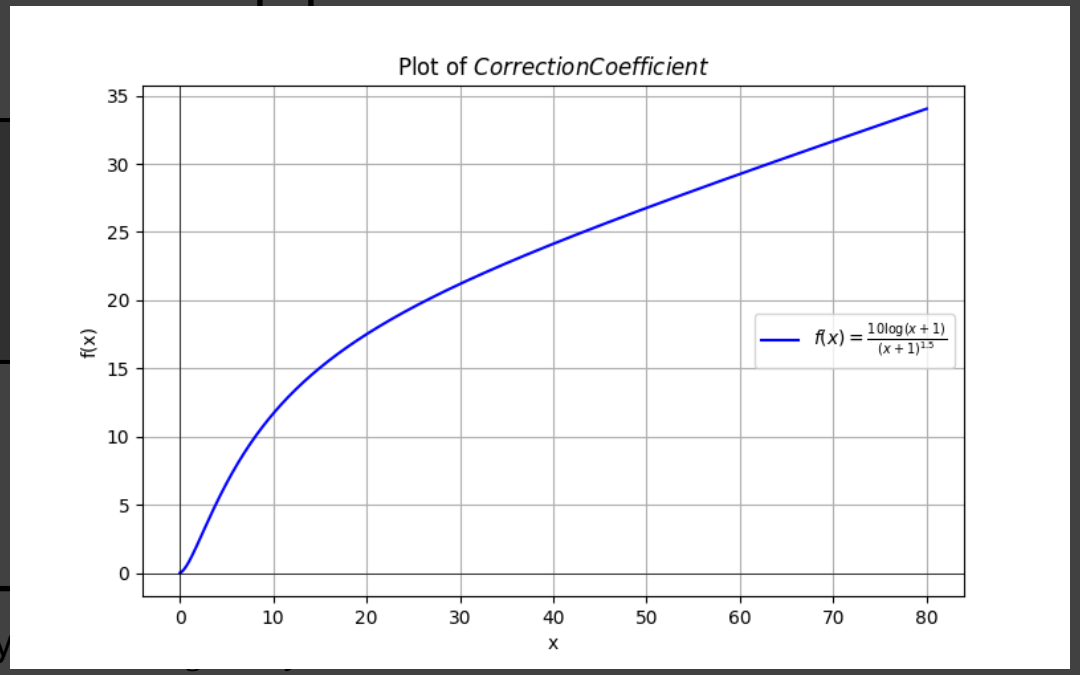
System Flow

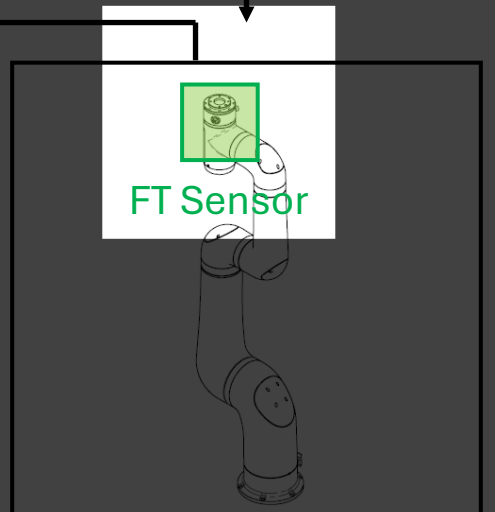
Force Feedback

Joint position (q)

Joint state + External wrench

Sim





Slave (Indy)

Gravity

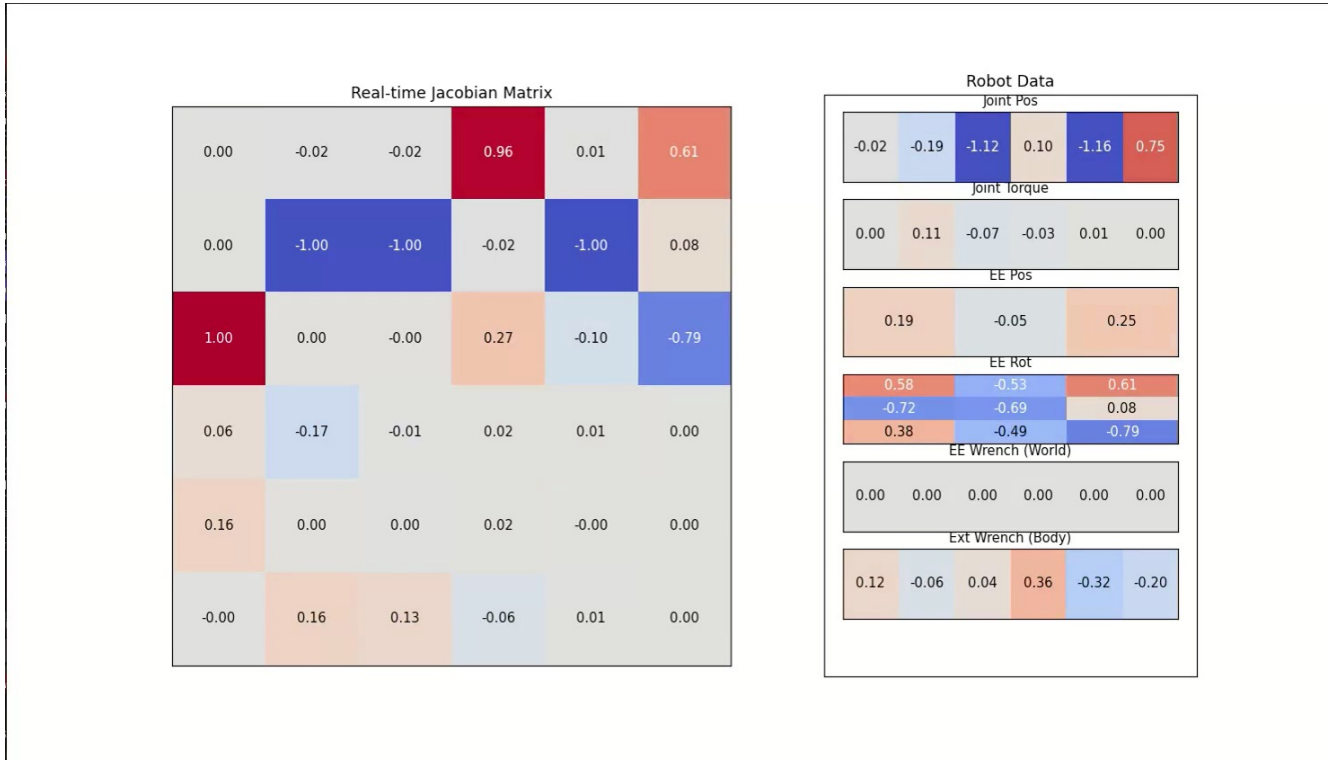
+ External Torque (τ_{ext})

$$\tau_{ext} = \mathcal{J}^T \mathcal{F}$$

System Flow

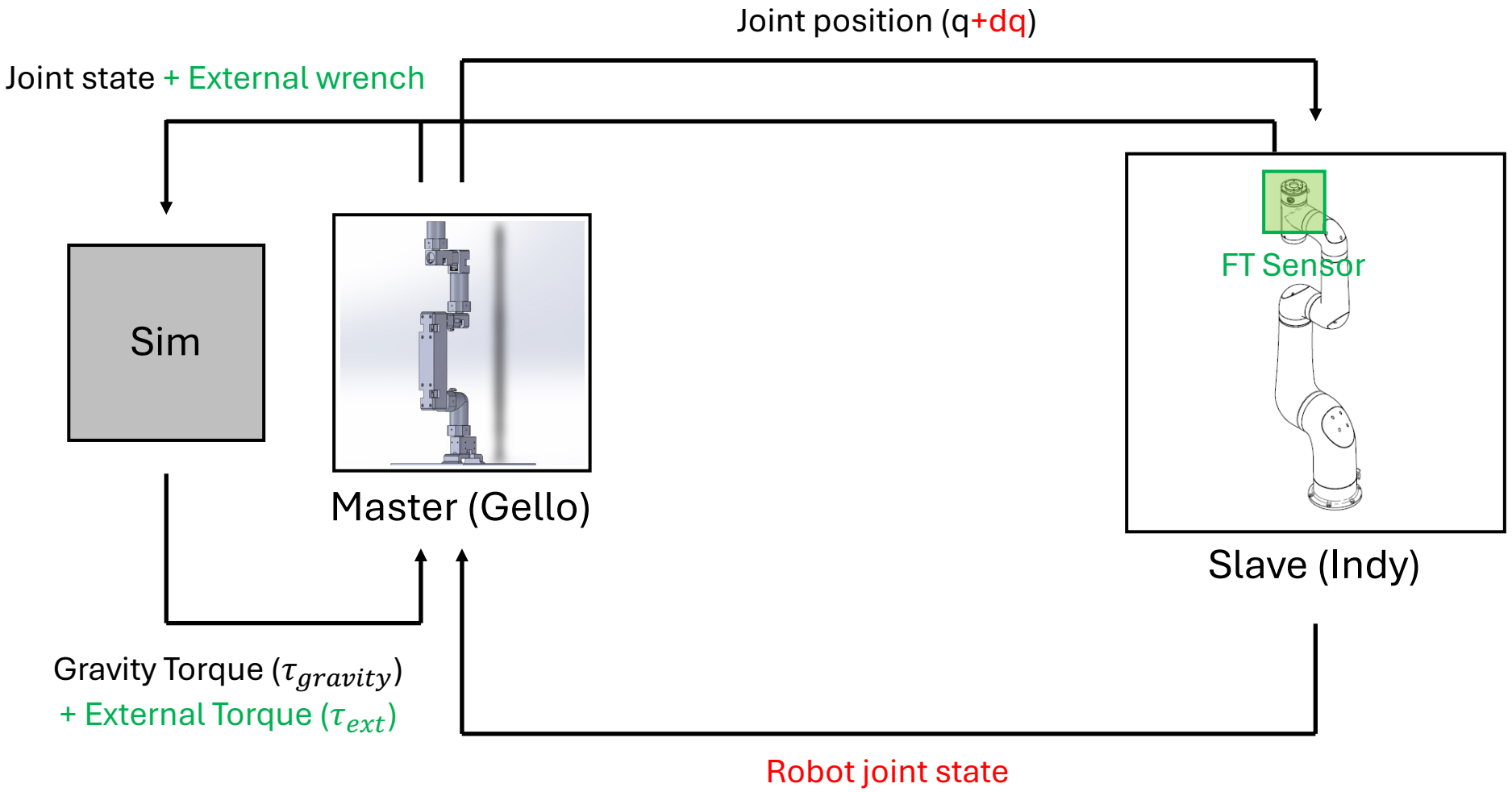
Force Feedback





System Flow

Compliance Control



System Flow

Compliance Control

- Robot deflects the trajectory based on the FT data from the force / torque sensor

- Robot follows adapted trajectory

>>> Safe control regardless of precise trajectory from the master

$$M\ddot{q}_d + C\dot{q} + K(q - q_0) = \tau_{ext} + \tau_{cmd}$$

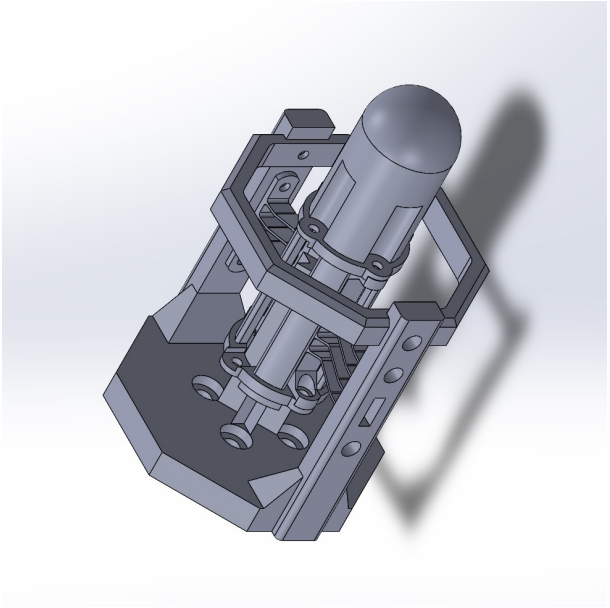


Compliance Tool

Miscellaneous Functinos



Source: Hackaday (3D Printed Blaster Does It With Compliant Components)





Gripper Device

Miscellaneous Functinos

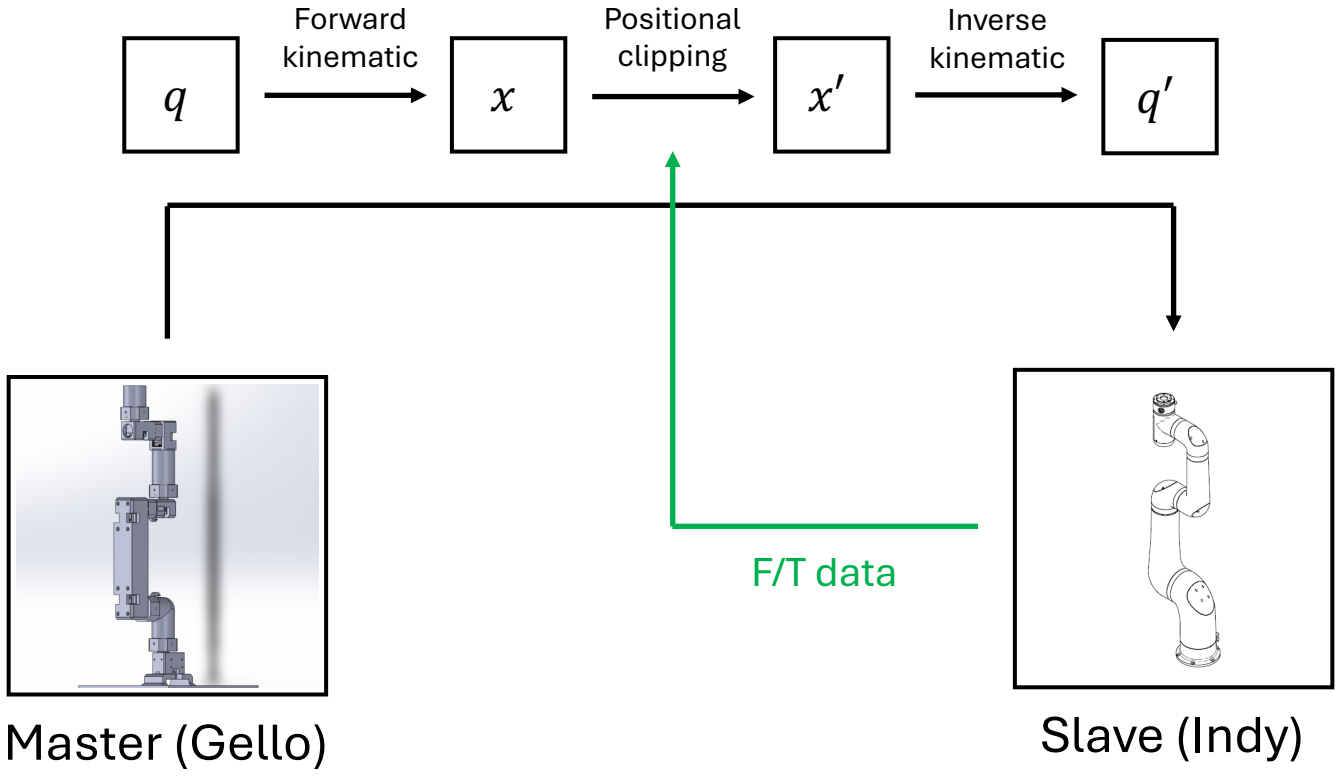


Tool Height Clipping

Miscellaneous Functinos

When f > threshold

$$\Delta x' = Orth_f \Delta x$$





Conclusion

1. Build joint control teleoperation system
   - Force mirroring, compliance control from the master side
   - Applicable for 6 dof robot arm with force torque sensor

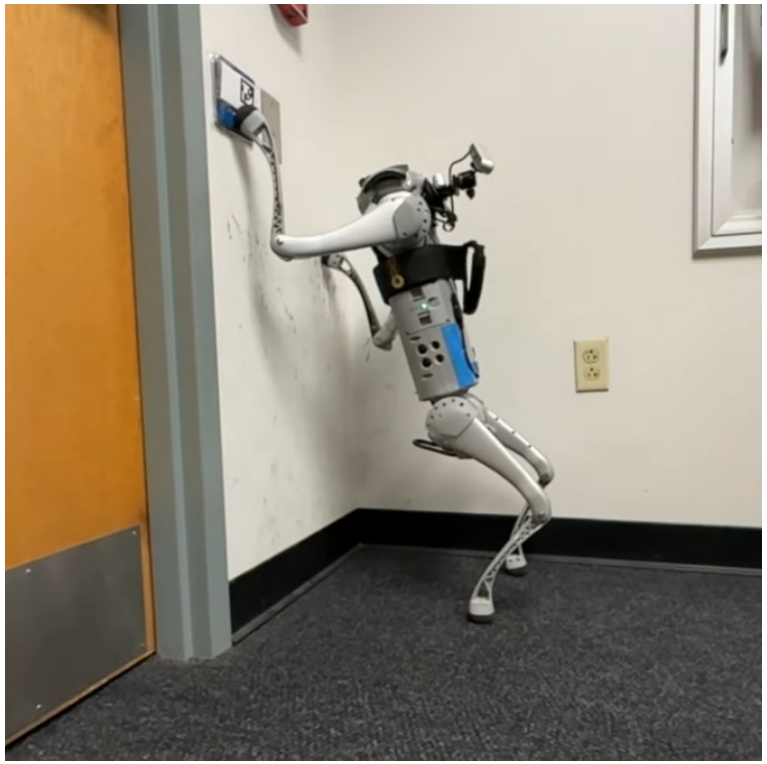
2. Learn types of control theories
   - Gravity compensation
   - Force feedback
   - Bilateral PD Control

3. Build the compliant end tool
   - 1D Impact dissipation

4. [Future Work] Use for data collection for imitation learning

Future Applications

- Contact involving manipulation
  - Point touch: Peg in hole, Pushing Button, 호떡 부치기, …
  - Linear touch: Brushing, Writing, Sanding, …
- Less stressful data collection



Source : IEEE (Robots Using Legs as Arms to Climb and Push Buttons)



Source : youtube (ABB's YuMi wishes you a Happy Lunar New Year)